CUSTOMER REPEAT PURCHASE & RETENTION ANALYSIS

An Olist Brazilian E-Commerce Case Study

Project overview | the full SQL analysis, statistical testing, and every chart referenced in this document live in the companion Jupyter notebook.

Prepared by	Data Analytics Team
Tools used	SQL (SQLite), Python, Pandas, Matplotlib, Seaborn, SciPy, Scikit-learn
Dataset	Olist Brazilian E-Commerce Public Dataset (Kaggle)
Observation period	2016 to 2018
Companion file	customer_repeat_purchase_retention_analysis.ipynb

Dataset source: https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

Executive Summary

Olist is a Brazilian e-commerce marketplace. Like most online retailers, it wants more customers to come back and buy again. This project analyzed three years of order history to answer a direct question: who comes back, and what separates them from customers who buy only once.

96,096	3.12%	28 days	R$302
Unique customers	Repeat purchase rate	Median time to a second purchase	Avg. spend, repeat customers

Headline findings

- Out of 96,096 customers, only 3.12% placed a second order during the period covered by the data. The remaining 96.88% were one-time buyers.
- Customers who did come back usually did so quickly. Half of all second purchases happened within 28 days of the first one.
- Repeat customers spent more overall (R$302 on average) than one-time customers (R$161), even though their individual orders were slightly smaller. The extra value comes from buying more than once, not from spending more per order.
- Review scores, delivery speed, customer location, and product category each showed some statistical relationship with repeat buying, but every one of these relationships was weak. A model built from all of them together could only correctly flag repeat customers about half the time (ROC-AUC 0.57).

Bottom line

Repeat purchasing at Olist is rare, and none of the factors available in this dataset (satisfaction, delivery, geography, category) explain it strongly on their own or in combination. The clearest lever available is timing. Since most repeat purchases happen within about a month, that first month after a purchase is the highest-value window for a follow-up offer or engagement effort.

Business Problem

Olist wants to understand why some customers return to buy again while most do not. This analysis investigates:

- How many customers are one-time buyers versus repeat buyers
- How often repeat customers purchase, and how quickly they come back
- Whether order value and total spending differ between the two groups
- Whether customer experience (reviews, delivery speed) relates to repeat buying
- Whether geography or product category relate to repeat buying
- Which of these factors, taken together, best explain repeat purchasing

The goal is to find patterns associated with repeat purchasing, not to claim that any single factor causes it. Correlation is not causation, and this distinction is treated carefully throughout the analysis, especially in the statistical testing sections of the companion notebook.

Customer Status Definitions

The dataset does not include a confirmed churn flag, so customer status is defined directly from observed order history. A **one-time customer** placed exactly one observed order. A **repeat customer** placed more than one. A customer is **observationally inactive** if no additional purchase occurred within the dataset's observation window. A one-time customer is not treated as a confirmed lost customer, since they may return after the dataset's 2018 cutoff date.

Dataset Overview

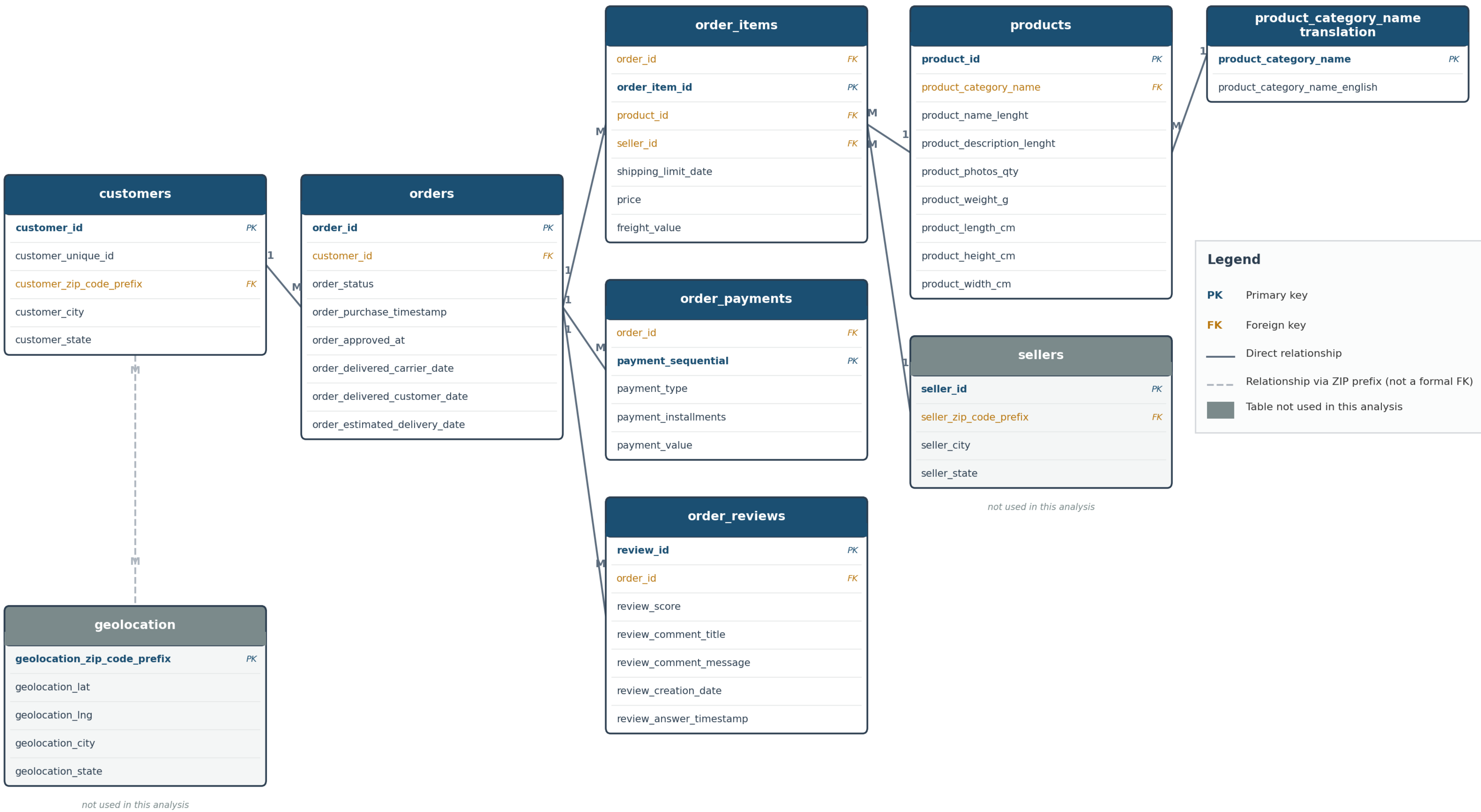
Item	Description
Dataset	Olist Brazilian E-Commerce Public Dataset
Business domain	E-commerce marketplace
Observation period	2016 to 2018
Customers analyzed	96,096 unique customers
Orders analyzed	99,441 orders
Source tables used	7 of 9 relational tables (full schema on the next page)
Database engine	SQLite, run in-memory through Python's sqlite3 module
Analysis languages	SQL for data retrieval, Python for statistics and visualization
Statistical methods	Welch's t-test, Mann-Whitney U test, Chi-square test, Cramer's V
Predictive method	Logistic regression
Customer identifier	customer_unique_id
Order identifier	order_id

The queries throughout this project run against SQLite for full reproducibility without an external database server. The SQL is standard enough to run on PostgreSQL, MySQL, or BigQuery with only minor syntax adjustments, mainly around date functions.

The full relational schema behind this dataset, including the two tables not required for this particular analysis, is diagrammed on the following page.

Olist E-Commerce Database Schema

9 tables. 7 were used directly in the customer repeat-purchase and retention analysis.



Analytical Approach

The analysis is organized as ten business questions, each answered primarily with SQL against the schema on the previous page, then interpreted with Python for statistical testing and visualization. Every query, chart, and statistical test referenced below is fully reproduced, with its exact SQL and output, in the companion notebook.

1. How large is the repeat-customer opportunity?
2. How often do repeat customers purchase?
3. Do one-time and repeat customers spend differently?
4. Are review scores associated with repeat purchasing?
5. Is delivery performance associated with repeat purchasing?
6. Does customer geography relate to retention?
7. Is a customer's first product category associated with repeat purchasing?
8. How long does it take customers to make a second purchase?
9. Does a combination of poor experience factors relate to lower retention?
10. Which characteristics, combined, best explain repeat purchasing?

Two populations, used deliberately

All observed orders (including cancelled or unfulfilled ones) are used for anything about *whether* a customer returns, since a customer's status as one-time or repeat should not depend on how their order was fulfilled. **Delivered orders only** are used for anything involving money (order value, spending), since those are the only orders that represent a completed, recorded sale.

A note on data quality and reproducibility

Several queries use window functions to identify each customer's first order. A small number of orders in this dataset share identical timestamps, and a small number have more than one review attached. Both situations can make a ranking non-deterministic unless the SQL includes an explicit, fully unique tiebreaker. Every ranking query in this project was written (and verified by re-running it) to return the same result on every execution.

Key Findings

96.88% of customers were one-time buyers.
Only 3.12% made a repeat purchase within the observed period.

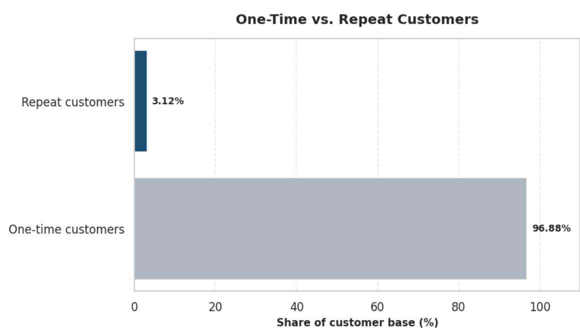
Repeat buying is shallow, not deep.
Among customers who did return, 9 in 10 stopped after exactly a second order.

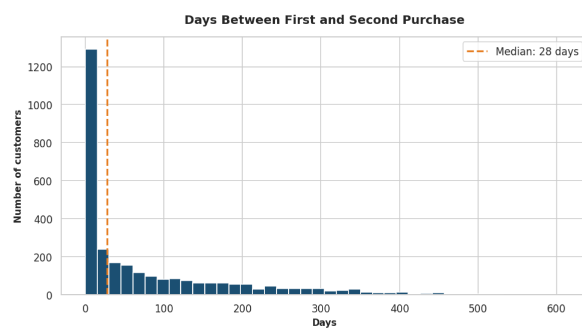
The median time to a second purchase was 28 days.
The average was much higher, about 80 days, because a smaller group of customers returned much later and pulled the average up.

Repeat customers are worth about twice as much over time.
They placed slightly smaller individual orders on average (R$148 vs. R$161) but spent roughly twice as much overall (R$302 vs. R$161), since their value comes from repetition rather than order size.

Experience, geography, and category all showed weak relationships at best.
Review score, delivery performance, and product category were statistically significant but practically weak (Cramer's V under 0.10 in most cases). Geography and the combined review-and-delivery experience did not clear the bar of statistical significance at all.

A combined predictive model performed barely better than chance.
A logistic regression using all available factors together reached an ROC-AUC of only 0.57, confirming this dataset does not contain a strong predictor of who becomes a repeat customer.





Two of the charts from the companion notebook. The full chart set, including every category, state, and statistical test breakdown, is in the notebook.

Business Recommendations

1. Focus on converting first-time buyers into second-time buyers.

This is where the largest opportunity sits, since the drop-off between a first and second purchase is the single biggest gap in the funnel.

2. Time follow-up offers around the first month after purchase.

The 28-day median time to a second purchase suggests this is the highest-value window for a reminder, discount, or engagement email.

3. Keep improving delivery reliability.

It is the one experience factor with a directionally consistent, if small, relationship to retention.

4. Do not rely on review score, delivery status, geography, or category alone to target retention campaigns.

None of them, individually or combined, reliably separates repeat customers from one-time customers in this data.

5. Do not put the current predictive model into production for customer targeting.

An AUC of 0.57 is not accurate enough to act on. Improving it would require richer behavioral data, such as browsing activity, marketing touchpoints, or pricing history, none of which is present in this dataset.

Limitations

- This analysis identifies associations, not causes. A statistically significant relationship between two variables does not mean one causes the other.
- Repeat customers make up only about 3% of the customer base. This imbalance limits how much any statistical test or model can learn about what makes them different, simply because there are relatively few of them to learn from.
- Customer status (one-time vs. repeat) is based only on orders observed within the dataset's 2016 to 2018 window. Some customers labeled one-time here may have purchased again after the data was collected.
- The dataset does not include marketing, pricing, or competitor information, all of which plausibly influence repeat purchasing but cannot be tested here.

Summary Table

Area	Result
Customer base	96,096 unique customers
One-time customers	93,099 (96.88%)
Repeat customers	2,997 (3.12%)
Average orders per customer	1.03
Median orders among repeat customers	2
Median time to second purchase	28 days
Avg. order value, one-time customers	R$ 160.65
Avg. order value, repeat customers	R$ 147.66
Avg. total spend, one-time customers	R$ 160.65
Avg. total spend, repeat customers	R$ 302.30
Review score finding	Significant, very weak association (Cramer's V = 0.042)
Delivery performance finding	Significant, very weak association (Cramer's V = 0.008)
Geography finding	Not statistically significant (p = 0.229)
Product category finding	Significant, weak association (Cramer's V = 0.069)
Combined experience finding	Not statistically significant (p = 0.072)
Predictive model	Logistic regression
Model accuracy / ROC-AUC	65% / 0.57
Model recall / precision (repeat class)	46% / 4%

Overall conclusion: repeat purchasing is rare and not well explained by the customer, experience, or product variables available in this dataset. Timing, specifically the first month after a purchase, is the clearest actionable lever this analysis identified.

For the full SQL behind every number in this document, every chart, and the complete statistical testing and modeling work, see the companion notebook: **customer_repeat_purchase_retention_analysis.ipynb**.